CSE400 — Fundamentals of Probability in Computing
Milestone 2 Scribe
Handover Probability in Drone Cellular Networks
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Scribe Question 1: Project System and Objective

Probabilistic Problem Definition

In Milestone Two, we formalize the probabilistic problem presented in Milestone One as a stochastic geometry-based problem and will analyse the stability of the association between a typical user equipment at the origin and the network of mobile Drone Base Stations (DBS) serving it. The crux of the mathematical representation of the problem is through the defining handover event, denoted as $H(t)$. Let $x_i(t)$ be the 3-dimensional position of the i-th drone at time t. The index of the serving drone at any instant in time, s, is given by the nearest-neighbour's rule for finding the closest drone.

$$S_s = \arg\min_{i \in N} ||x_i(s)||$$

The occurrence of handover at time t, referred to as $H(t)$ is defined as a change in the service index due to a change in serving base stations:

$$H(t) = \{\exists s < t : S_s \neq S(t)\}$$

Goal of the System

The primary objective is to derive an analytical model for calculating the Handover Probability. This probability is the cumulative distribution function (CDF) of the random variable representing the time until the first handover or the time between consecutive handovers. The group wishes to model the handover probability for two mobility models:

- **The Same Speed Model (SSM):** assumes that the speed of the Drones is constant and deterministic ($v_i = v$).
- **The Different Speed Model (DSM):** assumes that the speed of the Drones is a random variable ($V_i \sim f_V(v)$).

Sources of Uncertainty in the System

There are three sources of uncertainty in the system and have chosen to model them using stochastics:

1. **Initial Spatial Distribution:** The initial locations of the Drones are not deterministic (i.e., where the drone is when it first starts). The initial locations of drones are represented as a homogeneous Poisson point process (PPP) with a given intensity, λ_0 on 2D plane, and at a height h. Therefore, the number of Drones per area

A is represented as a Poisson random variable / Poisson Distribution:

$$P(N(A) = k) = \frac{(\lambda_0 A)^k e^{-\lambda_0 A}}{k!}$$

2. **Movement Direction (θ):** Drone movement: Each drone will independently fly along a linear path. The direction of movement to angle (θ) that it flies is uniformly distributed.

$$\theta \sim U[0, 2\pi)$$

   This introduces directional uncertainty i.e. any drone could go in any direction.

3. **Speed difference (DSM only):** The Different Speed Model gives different speed probabilities for each drone by using a PDF $f_V(v)$ (e.g., Rayleigh distribution, Uniform distribution). This creates uncertainty regarding the speed of the drones and thus also means that the handover probability must be integrated across the entire V domain as per lemma 3.

Scribe Question 2: Key Random Variables and Uncertainty Modeling

Drone Position ($\phi_D(0)$) and Nearest Neighbour Distance (u^*)

The Drone Base Station (DBS) position uses a homogeneous Poisson Point Process (PPP) to determine all positions of the DBS in the two-dimensional plane at height h, denoted by $\phi_D(0)$ with density λ_0. The fundamental random variable for this analysis is u^*, which is defined as the distance from the single user to the closest drone at $t = 0$, before use begins.

- **Assumption:** The number of drones $N(A)$ in any area A has a Poisson distribution as follows:

$$P(N(A) = k) = \frac{(\lambda_0 A)^k e^{-\lambda_0 A}}{k!}$$

- **Void Probability Derivation:** Void probability is probability of the region with area A in containing zero drones ($k = 0$):

$$P(N(A) = 0) = \frac{(\lambda_0 A)^0 e^{-\lambda_0 A}}{0!} = e^{-\lambda_0 A}$$

- **CDF Derivation:** The event that the nearest neighbour is greater than distance r ($P[u^* > r]$) equals the event that the circular area $A = \pi r^2$ around the user has no drones.

$$P(u^* > r) = P(N(\pi r^2) = 0) = e^{-\lambda_0 \pi r^2}$$

The CDF is the complement of this probability:

$$F_{u^*}(r) = P(u^* \leq r) = 1 - P(u^* > r) = 1 - e^{-\lambda_0 \pi r^2}$$

- **PDF Derivation:** The PDF is obtained by differentiating the CDF with respect to r.

$$f_{u^*}(r) = \frac{d}{dr} F_{u^*}(r) = \frac{d}{dr}(1 - e^{-\lambda_0 \pi r^2}) = 2\pi\lambda_0 r e^{-\lambda_0 \pi r^2}$$

This particular PDF $f_{u^*}(r)$ serves as the weight for the integral portion of the handover probability Equation (5) used in the Theorem 3 proof.

Direction (θ_i)

In this section, a mathematical approach will be used to derive the trajectory of each drone, which has been defined in straight line.

- **PDF:** drone's direction of movement, θ_i, is uniformly distributed random variable over the interval $[0, 2\pi)$. This uniform distribution will give a constant probability density function for each of the directionally randomly chosen drone movements, $f(\theta) = \frac{1}{2\pi}$.
- **Modelling:** The direction will be determined independently for each of the drone and therefore will be independent of the initial distances from the start position (i.e., the length and angle).

Speed (V_i) Distributions (DSM only)

With respect to the DSM, the speed V of the drones will be a statistically independent and identically distributed (i.i.d) random variable. The paper uses a specific probability function to model this uncertainty.

Rayleigh Distribution: (Used to model variance around a mean speed)

- **PDF:** $f_V(v) = \frac{v}{\sigma^2} e^{-v^2/2\sigma^2}, v \geq 0$
- **CDF:** $F_V(v) = 1 - e^{-v^2/2\sigma^2}$
- **Mean:** $E[V] = \sigma\sqrt{\frac{\pi}{2}}$

Uniform Distribution: (Used as an alternative bounded speed model)

- **PDF:** $f_V(v) = \frac{1}{b-a}, a \leq v \leq b$
- **CDF:** $F_V(v) = \frac{v-a}{b-a}$
- **Mean:** $E[V] = \frac{a+b}{2}$

The PDFs ($f_v(v)$), used in this section will be used in the integral equations that were previously described to determine the density of the non-serving drones and the final probability of handover.

Drone displacement Geometry (R)

Using vector geometry, the trajectory of a drone at time t relative to its original position is modelled.

- **Formula:** A drone that originally started off at a distance of r moves with speed v for time t, in the direction of θ. The new resulting distance (R) from the origin will be determined by using the Law of Cosines.

$$R(r, t, \theta) = \sqrt{r^2 + v^2t^2 - 2rvt\cos(\theta)}$$

The geometric formulation will be used directly in both Theorem 2 and Theorem 3 to define the dynamic limits of integration for computing the probability of handover.

Distance to serving drone ($u^*(t)$)

This variable describes how far away from the user's location is the nearest DBS at time 't'.

- **Modelling:** The locations of a DBS at two distinct times, t_1 and t_2, could not be obtained independently of one another. In the same speed model (SSM), computing distribution work due to network property preservation, leveraging the spatial equivalence proven in theorem 1. In contrast to this, in the discrete speed model (DSM), which is characterized by variable speeds, the temporal correlation that makes mathematically intractable, keep track of DBSs over time.

Non-Serving Drone Density ($\phi'_D(t)$) in DSM

In the DSM, the point process $\phi'_D(t)$ representing all drones in the air with the exception of the serving drone is modelled as an inhomogeneous Poisson point process (PPP).

- **Formula:** Because the user is connected to the nearest drone at time $t = 0$, there exists a temporary zone around the user in which the density is 0. The user will have created a density function $\lambda(t; u_x, u^*)$ at a distance of u_x from user location as a result of integrating the speed probability density function, $f_V(v)$, as discussed in Lemma 3:

$$\lambda(t; u_x, u^*) = \lambda_0 \left[1 - \int_{\frac{u^*-u_x}{t}}^{\frac{u^*+u_x}{t}} f_V(v) \frac{1}{\pi} \cos^{-1} \left(\frac{v^2t^2 + u_x^2 - (u^*)^2}{2vtu_x} \right) dv \right]$$

Equation (4) demonstrates the manner in which the temporary exclusion zone that is created by the user distorts the drone density as time progresses. As $t \to \infty$, the point process for non-serving drones will have been completely intermixed (conditioned) and the density will converge to λ_0.

How Uncertainty is Modelled: Why These Distributions?

Stochastic geometry is utilized by the researchers to accommodate both the spatial and temporal uncertainties of the network.

- **PPP:** A Poisson Point Process (PPP) is mathematically tractable and can be used to provide an accurate match for uncoordinated random deployments. The most important aspect of the PPP is that it allows for the use of the Displacement Theorem (Lemma 1), which guarantees that the λ_0 density will be preserved and that these drones will continue to form a homogeneous PPP as they each move independently.

- **Rayleigh Speed:** The majority of drones will fly with an average operational speed; however, there will be a significantly smaller number of drones that are operating near either an extreme low or an average operational speed. This speed distribution is well modelled using the Rayleigh curve.

- **Uniform Direction:** Out of all DBSs, since there is no coordination, all possible flight directions are equally likely. Therefore, the resulting flight direction distribution of DBSs can be said to follow a uniform distribution.

- **Note:** The PDF of the nearest neighbour distance (u^*) is, in form, very similar to a Rayleigh distribution. This is strictly a geometric consequence of the PPP void probability and is not a result of an independent agreement of previous distributional characteristics.

Scribe Question 3: Probabilistic Reasoning and Dependencies

Spatial and Kinematic Independence (The Displacement Theorem)

To retain analytical tractability, it is necessary for our mathematical model to be based on the assumption of independence. The initial spatial distribution of Drone Base Stations (DBSs) will be an independent Poisson Point Process (PPP). It is also assumed that each DBS makes its choice of straight-line trajectory and direction completely independently from the other DBSs. The paper uses strict independence to apply the Displacement Theorem (or Lemma 1). According to this theorem, if points of a Homogeneous PPP are displaced independently (with identically distributed displacements), the final configuration will be a Homogeneous PPP with the same density, λ_0, as before. The probabilistic calculation is important, since it ensures that for any future time, t; mobile drones in the Same Speed Model (SSM) will have the same characteristics of the original network and thus can produce geometrically consistent calculations.

Conditionality for the System Integration

The primary mathematical technique to determine the probability formula for handover is through conditionality. Instead of trying to compute the complicated total probability directly, the system computes the probabilities of handover based on previously determined geometric values. The SSM computation for the (Theorem 2), For an initial serving distance of 'r', to discover the total probability that there will be no handover from an initial location to a target location, if you move along an angle θ, it can be expressed as the conditional handover probability where $(C_2 \setminus C_1)$ are the newly created circular formations due to the direction (θ). Thus, the SSM model can calculate the probability of the drone making a handover through the following mathematical relationship:

$$\mathbb{P}[H(t) \mid r, \theta] = 1 - \mathbb{P}[N(C_2 \setminus C_1) = 0] = 1 - e^{-\lambda_0(\pi R^2 - S)}$$

Where S is the exact size of the mathematical area of overlap between the original serving area (C_1) and the new offset serving area (C_2). After the SSM model has been able to establish the SSM relationship between the two variables the final outcomes must be determined by "Deconditioning" the values of these two relationships. To achieve this, the SSM model must compute the unconditional probability of the SSM relationship. This is done by employing all of the values of r and θ and then integrating the values of r and θ according to the Probability Distribution Functions that were established by the model's system (i.e., Equation (2)) to provide the model with a value for the unconditional probability of the SSM relationship.

Temporal Dependence and the DSM Lower Bound

In summary, this paper indicates that there exists an exact probabilistic dependency that arises in terms of the DSM's conceptual framework. Although the model supports the premise that the actions of the drones are independent from one another when travelling through space, the position of each drone on a given time (t_2) has a very high probability of being located adjacent to its original position (t_1) based on its respective operating speeds. As a result of the methodology used to describe both events (no specific formulae exist for this temporal correlation) an exact mathematical formula for determining adjacent neighbour locations is geometrically impractical. However, with the use of this temporal correlation through Theorem 3, this model produces a Lower Bound. Through the definition of event G (the potential for no DBS to exist in the bounding circle C_2) and the establishment of the event 'no handover' event $H'(t)$ as a subset of G ($H'(t) \subset G$), i.e. 'no handover' occurs at the boundary of every event $H'(t)$ due to the relative speeds of the aircraft at the boundary causing the potential for drones to have entered or exited C_2 at time t, $\mathbb{P}[H'(t)] \leq \mathbb{P}[G]$ is used.

$$\mathbb{P}[H(t)] \geq 1 - \mathbb{P}[G]$$

which forms the foundational basis for the complex integral evaluated in Equation (5).

Equivalence Through Translation Invariance

Finally, the model uses a powerful probabilistic equivalence (Theorem 1) to bridge aerial and terrestrial networks. It proves mathematically that the handover probability for a static user facing mobile drones is exactly equivalent to that of a mobile user moving through static terrestrial base stations. This reasoning is strictly supported by the translation invariance property of a homogeneous PPP. Mathematically, shifting a stationary PPP (Φ_B) by a movement vector $x(t)$ result in a relative point process ($\Phi_B - x(t)$) that is fundamentally indistinguishable from the original PPP. Because Φ_B is a homogeneous PPP, it is translation invariant, yielding the equivalence $\tilde{\Phi}_D(t) \equiv \text{PPP}(\lambda_0)$.

Scribe Question 4: Model-Implementation Alignment

Question: Explain how the theoretical probabilistic model is implemented in practice (e.g., via simulation). Discuss the parameters, results, and the key mathematical assumptions that influence the system design.

1. Simulation Approach (Monte Carlo Methodology)

The theoretical probabilistic model is practical implemented through Monte Carlo simulations. The variables defined in the system model are translated into a step-by-step algorithmic execution:

1. **Step 1 - Initialization (Spatial Statistics):**

   To simulate the Homogeneous Poisson Point Process (PPP), the implementation first generates the total number of Drone Base Stations (DBSs) in a bounded simulation area using a Poisson random variable:

   $$N \sim \text{Poisson}(\lambda_0 \times \text{Area})$$

   These N drones are then assigned uniform x and y coordinates. Initial directions are assigned as:

   $$\theta_i \sim U[0, 2\pi)$$

   and speeds V_i are assigned based on the chosen PDF $f_V(v)$ (constant for SSM, random for DSM).

2. **Step 2 - Time Evolution (Kinematics):**

   The mathematical distance evolution is computationally implemented by updating the position vectors of all drones linearly over time. The new position of the i-th drone at time t is calculated as:

   $$x_i(t) = x_i(0) + V_i t[\cos(\theta_i), \sin(\theta_i)]$$

   At each time step:

   - The Euclidean distance from the origin to all drones is calculated.
   - The minimum distance is identified.

- The index of the serving drone is determined.
- A handover is recorded if this serving index changes from its $t = 0$ state.

3. **Step 3 - Probability Estimation:**

   The mathematical expectation $\mathbb{P}[H(t)]$ is estimated empirically over thousands of independent iterations as the ratio:

$$\mathbb{P}[H(t)] \approx \frac{\text{Number of trials with a handover by time } t}{\text{Total number of Monte Carlo trials}}$$

2. Parameter Alignment

The simulations use specific numerical parameters to validate the mathematical integrals (Equation 2 and Equation 5). The paper aligns the models using:

- Density: $\lambda_0 = 1$ DBS/km^2
- Speed (SSM): A deterministic velocity of $v = 45$ km/h
- Speed (DSM): A random variable V with a mean speed of $\mathbb{E}[V] = 45$ km/h to provide a fair comparison against the SSM
- Time Window: Observations are recorded over an interval of $t = 0$ to 100 seconds

3. Results Alignment and Bound Tightness

- **SSM Validation:**

  At $t = 100$ seconds, both the exact analytical expression (Theorem 2) and the Monte Carlo simulation yield a handover probability of:

$$\mathbb{P}[H(t)] \approx 0.40$$

  The perfect alignment confirms the validity of the equivalence principle (Theorem 1).

- **DSM Bound Tightness:**

  Because the DSM relies on a lower bound (Theorem 3) rather than an exact formula, the simulation serves to test the "tightness" of this bound.

  At $t = 100$ seconds:

$$\text{Simulation} \approx 0.35$$

$$\text{Analytical lower bound} \approx 0.34$$

This narrow gap proves that the derived mathematical bound is highly tight and serves as an accurate proxy for the true probability.

4. Key Modeling Assumptions Influencing Implementation

- **Nearest-Neighbor Policy:** This simplifies the handover event strictly to a geometric calculation of distance arrays. A handover geometrically corresponds to a drone crossing the perpendicular bisector (Voronoi cell boundary) between the current server and the candidate server.
- **Straight-Line Mobility:** Assuming

  $$x_i(t) = x_i(0) + \vec{v}t$$

  allows the implementation to use simple linear algebraic updates rather than simulating complex trajectory algorithms.
- **Constant Height (h):** By assuming all drones fly at the exact same altitude h, the 3D Euclidean distance calculations reduce to 2D planar distances plus a constant h^2. The researchers project the user vertically to the drone plane (o'), allowing the simulation to operate strictly in 2D space without loss of accuracy.

Scribe Question 5: Cross-Milestone Consistency and Change

Question: Reflect on how your understanding of the probabilistic model has evolved from Milestone 1 to Milestone 2. What is now well-defined, and what still needs clarification?

1. Evolution of the Poisson Point Process (PPP)

Milestone 1 State: The group initially understood the Homogeneous Poisson Point Process (PPP) strictly as a conceptual tool to represent randomly placed drones without coordination.

Milestone 2 Refinement: The mathematical mechanics of the PPP are now fully integrated into the model. The group understands that the PPP allows for the calculation of the void probability, which is mathematically derived to formulate the nearest-neighbor distance Probability Density Function (PDF):

$$f(r) = 2\pi\lambda_0 r e^{-\pi\lambda_0 r^2}$$

This serves as the foundational weighting function for all subsequent spatial integrations in the paper.

2. Justification of the Equivalence Principle

Milestone 1 State: Theorem 1 was viewed as a conceptual observation.

Milestone 2 Refinement: The equivalence is mathematically grounded in the translation invariance property of a homogeneous PPP. Shifting all points in a stationary PPP (Φ_B) by a vector $x(t)$ preserves the exact distribution of the PPP. Therefore, the relative geometric calculations remain identical.

3. Mathematical Breakdown of the Different Speed Model

Milestone 1 State: DSM only provided a lower bound without clear probabilistic reasoning.

Milestone 2 Refinement: The varying speeds break the spatial similarity transformation present in SSM. This creates temporal dependence, causing the non-serving drone locations ($\Phi'_D(t)$) to distort into an inhomogeneous PPP. The lower bound is defined via the time-dependent density function:

$$\lambda(t; u_x, u^*)$$

4. What is Now Well-Defined

- Exact derivations of CDFs and PDFs for drone distance and speed.
- Application of the Displacement Theorem (Lemma 1).
- Theorem 2:

$$\mathbb{P}[H(t)] = \int_0^\infty \int_0^{2\pi} \left(1 - e^{-\lambda_0(\pi R^2 - \mathcal{S})}\right) f_\theta(\theta) f_{u^*}(r) d\theta dr$$

Where:

$$f_\theta(\theta) = \frac{1}{2\pi}$$

$$f_{u^*}(r) = 2\pi\lambda_0 r e^{-\pi\lambda_0 r^2}$$

5. What Still Needs Clarification

- Numerical methods required to evaluate nested integrals.
- Step-by-step analytical derivation of Lemma 3.

Scribe Question 6: Open Issues and Responsibility Attribution

Question: Detail the remaining open issues, unresolved probabilistic questions, and assign specific responsibilities to group members for the next milestone.

1. Resolved Probabilistic Modeling (M1 to M2)

- Justification for using Homogeneous PPP via Displacement Theorem (Lemma 1).
- Validation of Theorem 1 via translation invariance.
- DSM lower bound defined via inhomogeneous density function $\lambda(t; u_x, u^*)$.

2. Open Mathematical and Analytical Issues

- Derivation of Lemma 3 (trigonometric terms in Equation 4).
- Independent verification of numerical integration in Theorem 2 and Theorem 3.
- Investigation of strict upper bounds for DSM.
- Quantification of tightness as a function of λ_0 and speed variance σ^2.

3. Responsibility Attribution for Milestone 3

Entire Group:

- Study Monte Carlo simulation convergence.
- Review validation plots.

Scribe:

- Document full derivations of Lemma 2 and Lemma 3.

Mathematical Analysis:

- Verify numerical integration of Equation (5).
- Match analytical and simulation distributions for different $f_V(v)$.

GitHub Management:

- Organize simulation codebase.